

ARGUS: Auditing Causal Identification Credibility in Empirical Difference-in-Differences Studies

Yonghong Zhang

Departamento de Financiación e Investigación Comercial,
Universidad Autónoma de Madrid, Madrid, Spain
`yonghong.zhang@estudiante.uam.es`

Abstract. Empirical economics leans heavily on difference-in-differences (DID) designs, whose credibility rests on identification assumptions that are often asserted rather than audited. We present **ARGUS**, a responsible-AI framework that audits the *identification credibility* of such studies—whether their reported evidence adequately supports each identification assumption—instead of adjudicating the true causal effect, which is never observed. ARGUS decomposes identification into auditable dimensions, gathers in-paper evidence, assesses each assumption–implication–evidence chain, and localizes risk in a transparent, human-reviewable report. We validate the auditor with synthetic flaw injection, which supplies reproducible local ground truth without the true effect: a deterministic keyword baseline detects only 0.182 of injected flaws because it cannot see flawed-but-present evidence, whereas judging evidence *adequacy* catches the flaws keyword presence misses (1.000 detection, no false alarms). Presence, in other words, is not adequacy. On 27 real DID papers the binding constraint shifts to evidence grounding rather than over-flagging. A pilot human-gold study then exposes a systematic over-severity bias that a lightweight, anchor-based calibration layer reduces while preserving high-risk recall. ARGUS is thus a high-sensitivity screening tool that flags dimensions for expert review rather than issuing verdicts; the human-gold study is pilot-scale.

Keywords: Responsible AI · Causal inference · Difference-in-differences · Environmental policy · LLM agents.

1 Introduction

Difference-in-differences (DID) is one of the most widely used research designs in empirical economics, and a particularly visible one in environmental policy, our motivating case. Studies of carbon pricing, emissions-trading pilots, and industrial-emission standards routinely compare treated and untreated units before and after a policy to recover its causal effect. The credibility of any such estimate rests entirely on a chain of *identification assumptions*: that treated and control units would have moved in parallel absent the policy, that units did not

anticipate it, that staggered adoption is handled with an appropriate estimator, that there is no interference between treated and control units, and so on. A decade of methodological work has shown how fragile each of these assumptions can be, and how easily a standard two-way fixed-effects specification can mislead under heterogeneous and staggered treatment timing [7,4,19,5,16].

Yet in applied work these assumptions are frequently *asserted* rather than *audited*. A paper states that parallel trends “hold”, shows an event-study plot, and moves on; a reader who wants to know whether the identification argument is actually supported by the reported evidence must reconstruct it by hand, dimension by dimension. This is slow, expert-intensive, and inconsistent across reviewers. The result is a reliability gap in a body of evidence that increasingly informs environmental policy.

Our stance. We argue that identification *credibility* is auditable even though the causal *effect* is not. In a DID design the counterfactual is never observed, so asking an automated system whether the estimated effect is true is both hopeless and, we think, the wrong target. We ask instead a question that can be grounded in the paper itself: *is each identification assumption supported by the evidence the study reports, and where is that support weakest?* This reframing turns an unanswerable causal question into a tractable auditing problem over assumptions, their testable implications, and the in-paper evidence that bears on them. A useful system should therefore localize where support is fragile rather than issue a verdict.

Why human calibration. Identification credibility is not a binary truth label. A dimension is rarely simply “satisfied” or “violated”; reviewers grade it on a severity scale, and where the low/medium/high boundaries fall is a matter of expert judgement. Some disagreement is therefore intrinsic, especially when a DID-specific check is applied to a non-standard design where the dimension may not apply as conventionally defined. We treat this ambiguity as a feature of the task, not noise: ARGUS pairs automated, evidence-grounded auditing with a *human-calibrated severity* scale, using a pilot expert gold both to *evaluate* the auditor and to *calibrate* how severely it flags.

ARGUS. We introduce **ARGUS**, a responsible-AI framework that audits the identification credibility of DID environmental-policy studies. ARGUS decomposes identification into a fixed set of auditable dimensions, gathers the supporting evidence each dimension requires, assesses every assumption–implication–evidence chain, localizes risk across dimensions, and produces a transparent, evidence-anchored report for a human expert (Figure 1). It is a *bounded-agentic* system by design: the pipeline between modules is fixed and deterministic, and model-driven agency is confined to two scoped sub-tasks. Because the true effect is unavailable as ground truth, we validate the auditor with *synthetic flaw injection*: we perturb a credible paper with one known identification flaw and check whether ARGUS detects it, stays quiet on the clean version, and localizes

the right dimension. The injected flaw is local ground truth that requires no access to the true effect. Deploying ARGUS on real papers then exposes *evidence grounding* (retrieving the right evidence) as the binding constraint, and a pilot human gold reveals a systematic over-severity bias that a lightweight, anchor-based calibration layer reduces while preserving high-risk recall. ARGUS is thus best read as a high-sensitivity screening tool that flags dimensions for expert review rather than an autonomous arbiter of causal truth.

Contributions. This paper, presented as a doctoral-research plan, makes four contributions. (i) a structured **11-dimension audit framework** for DID identification credibility whose unit of audit is an explicit *assumption-implication-evidence* chain, targeting reported-evidence adequacy rather than causal truth (§3); (ii) a **bounded-agentic** evidence-retrieval and assessment pipeline, with model agency confined to two scoped, traceable sub-tasks behind a fixed deterministic control flow; (iii) a **multi-stage evaluation** built on *flaw injection as local ground truth* for identification auditing (it needs no access to the true effect), showing that adequacy reasoning catches flawed-but-present evidence that keyword presence cannot, with a real-paper deployment on 27 DID papers that relocates the bottleneck to *evidence grounding* and separates retrieval failure from substantive high risk; an *architecture ablation* against a same-model single-pass baseline shows the bounded design raises synthetic flaw detection from 0.455 to 1.000 and, on real papers, preserves an explicit **unknown** state (38.9% vs 0.4%) that keeps evidence-grounding uncertainty visible rather than committing a guess (§5); and (iv) a **pilot human-gold calibration loop** (5 papers, 55 dimension-level cells) in which disagreement is largely design-applicability-driven: comparing ARGUS to an adjudicated gold reveals a one-directional over-severity bias (more severe than gold in 28/33 answered cells, less severe in none), and an applicability-aware, anchor-based calibration layer reduces it while preserving high-risk recall. Expert annotation can therefore both evaluate and calibrate an identification audit. The human-gold analysis is pilot-scale; larger expert-labelled corpora are future work.

2 Related Work

DID identification and its threats. A large recent econometrics literature has formalized when DID estimators fail. Goodman-Bacon [7] decomposes the two-way fixed-effects estimator under variation in treatment timing and exposes the “forbidden” comparisons it makes between newly and already-treated units; de Chaisemartin and D’Haultfœuille [5] show such estimators can place negative weights on treatment effects; and Callaway and Sant’Anna [4] and Sun and Abraham [19] propose estimators robust to heterogeneous, staggered effects. Roth et al. [16] survey the design’s assumptions and the subtleties of pre-trend testing, while Bertrand et al. [1] show that DID *inference* is itself fragile once serial correlation is ignored. ARGUS does not add an estimator; it *operationalizes* this literature as an auditable rubric, turning each known threat into a dimension whose supporting evidence can be checked in a given paper (§3).

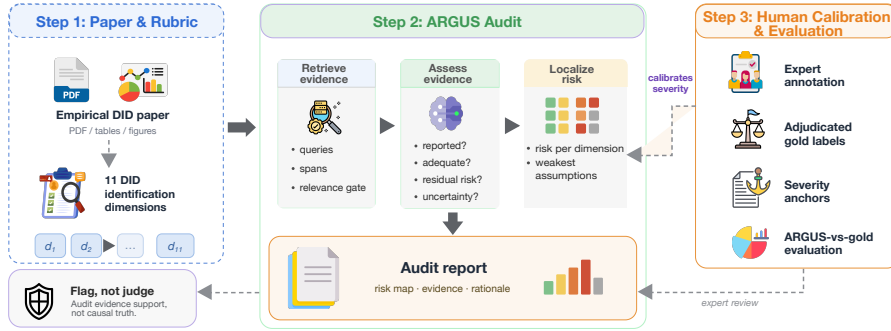


Fig. 1. Overview of ARGUS. Step 1 converts an empirical DID paper into an auditable structure using an 11-dimension identification rubric. Step 2 retrieves dimension-specific evidence, assesses reported-evidence adequacy with a bounded LLM audit module, and localizes risk in a human-reviewable report. Step 3 uses expert annotation and adjudicated gold labels to evaluate and calibrate audit severity. ARGUS flags dimensions for expert review; it does not adjudicate causal truth.

LLM causal reasoning and benchmarks. A parallel line asks whether large language models can reason about causality at all. Benchmarks such as CLadder [10], Corr2Cause [9], and CausalBench [25], together with broader surveys and critiques [11,20], find that models often reproduce causal *language* without reliable causal *inference*—“causal parrots” that talk causality but do not perform it [22]. Most such benchmarks score abstract or synthetic causal queries; a few move toward applied settings, including context-aware causal-direction judgement in social science [13] and the disentangling of causal *identification* from *estimation* [17]. ARGUS shares this last distinction but inverts the task: rather than asking a model to produce a causal answer, it audits whether a *human* paper’s identification assumptions are adequately evidenced, and abstains from a causal verdict by design.

Responsible AI for empirical research. A growing body of work applies AI to research reliability—automated reproducibility checking, statistical-reporting validation, and AI-assisted peer review. The need is real: when many independent teams analyse the same data, defensible analytic choices alone produce a wide spread of results [18,2], and causal work in economics shows evidence of specification searching and publication bias [3]. ARGUS is complementary but differs in target and in epistemic posture. First, it audits *identification credibility* specifically, the part of an empirical argument that determines whether a causal claim is licensed at all, rather than surface statistical reporting. Second, it does not issue verdicts: it produces evidence-anchored, human-reviewable reports that localize risk, leaving adjudication to a domain expert. This human-in-the-loop, flag-not-judge stance follows from the counterfactual being unobserved: automated causal verdicts would be both unreliable and unaccountable.

Benchmark and evaluation methodology. Our evaluation design follows three lessons from the benchmarking literature. Holistic evaluation [14] argues against collapsing capability to a single accuracy number; ARGUS likewise reports a layered profile (retrieval quality, relevance, adequacy, and localized risk) rather than one detection rate. Dynamic, adversarial benchmarking [12] warns that static benchmarks saturate; our flaw-injection loop is an adversarial, model-in-the-loop construction that we intend to escalate as the assessor improves (our synthetic flaws already saturate, motivating harder ones). Finally, Raji et al. [15] caution that turning an abstract capability into a measurable task is a consequential, non-neutral move; we take this seriously in §6, where we state what ARGUS does and does not measure. Should the annotated corpus be released, we will document it with a datasheet [6].

Execution-grounded verification vs. assumption auditing. A complementary line asks whether a model can *carry out* causal inference correctly. CausalVerify [23], an execution-grounded benchmark, runs a model’s generated estimation code against synthetic data and checks that the recovered effect matches the canonical estimator on the realised dataset. ARGUS audits a different layer: not whether the estimation is *executed* correctly, but whether the identification *assumptions* behind it are adequately supported by the paper’s reported evidence—which that benchmark explicitly places outside its scope. ARGUS builds on the corpus of published economics papers released with that benchmark [24], adding a new dimension-level identification-risk annotation layer on top of it.

Bounded LLM agents. Tool-using agents in the ReAct style interleave reasoning and acting over an open horizon [21]. ARGUS deliberately *bounds* this paradigm. Agency is confined to two scoped sub-tasks (extraction and assessment), each run as a fixed-tool, hard-step-budget loop with a logged trace, while the inter-module control flow is ordinary deterministic code. We trade autonomy for two properties the framework depends on: *reproducibility*, so that the flaw-injection metrics are stable and comparable across runs, and *evidence-traceability*, so that every flag a human reviews is anchored to inspectable evidence. For responsible auditing we regard this restraint as appropriate rather than limiting.

3 The ARGUS Framework

3.1 Identification rubric

ARGUS decomposes a DID study into eleven auditable dimensions: parallel trends, no anticipation, staggered-timing handling, SUTVA/spillovers, control-group construction, specification, inference, sample-period selection, concurrent policies, placebo/robustness, and data measurement. Each dimension is expressed as an *assumption* (what must hold for identification), a testable *implication* (what should therefore be observable), and the *evidence* a credible paper would report to support it—for example, the parallel-trends dimension expects an event-study or pre-trend test with null pre-period coefficients. The rubric is

declarative, stored in a configuration file (`identification_dimensions.yaml`), so that the set of audited dimensions is explicit and inspectable rather than buried in code. A companion flaw taxonomy (§4) maps each known identification threat one-to-one onto a dimension, which both documents the rubric’s coverage and supplies the perturbations used for evaluation.

3.2 Bounded-agentic pipeline

The audit runs a fixed, deterministic sequence—*decomposition* $\rightarrow$ *extraction* $\rightarrow$ *assessment* $\rightarrow$ *localization* $\rightarrow$ *report*—shown in the top band of Figure 2. The stage order is written in code and is never chosen by a model. *Decomposition* instantiates the rubric for the paper at hand. *Extraction* gathers, per dimension, the in-paper evidence relevant to that assumption. *Assessment* evaluates each assumption–implication–evidence chain and emits a per-dimension risk judgement with a rationale and cited evidence. *Localization* aggregates these judgements into a risk map that ranks where support is weakest. *Report* renders a transparent, evidence-anchored document for a human reviewer.

Agency is confined to the two stages where open-ended judgement actually helps, extraction and assessment. Each invokes a bounded ReAct-style loop with three constraints: (i) a *small fixed tool set* passed in per call—`evidence_search` over the paper’s text, figures and tables, `figure_parse` for structured figure/table content, and `policy_lookup` for policy metadata such as announcement versus implementation dates; (ii) a *hard step budget* (`max_steps`) so a run cannot wander indefinitely; and (iii) a *logged trace* of every step, tool call, and cited piece of evidence. All other stages are plain orchestration with no agent loop.

Why bound the agency. A fully autonomous agent would make per-run behaviour path-dependent and high-variance, which would undermine the central evaluation claim of §4—that flaw injection provides *reproducible* local ground truth. Restricting agency to two clearly scoped sub-tasks preserves reproducibility (stable, comparable metrics) and evidence-traceability (every judgement anchored to logged evidence a human can inspect). The expert sits at the end of the pipeline: ARGUS surfaces and localizes identification risk, it does not adjudicate it.

3.3 Output

ARGUS emits (*study*, *identification dimension*, *risk*) triplets, each with an evidence-grounded rationale, together with the human-readable report. It never produces a verdict on the true causal effect. This output shape is what makes the system both auditable by a human and measurable by the evaluation that follows.

4 Evaluation by Flaw Injection

4.1 Local ground truth without the counterfactual

The defining difficulty in evaluating a causal-identification auditor is that the quantity of ultimate interest, the true causal effect, is never observed. We there-

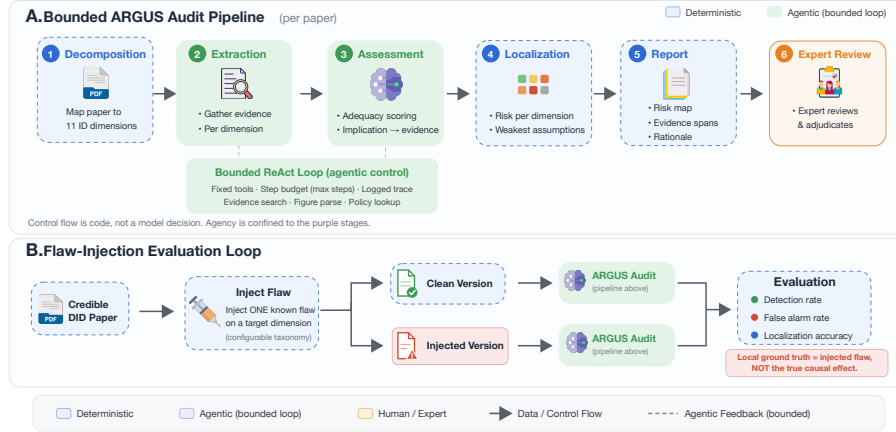


Fig. 2. ARGUS architecture. The top band is the audit pipeline, a fixed deterministic sequence; agency is confined to the two agentic stages (extraction and assessment), each a bounded ReAct loop with a fixed tool set (evidence search, figure parse, policy lookup), a hard step budget, and a logged trace. The bottom band shows the flaw-injection evaluation loop, which injects one known flaw, audits the clean and injected versions, and scores detection, false alarm, and localization against the injected flaw as local ground truth—never against the unobserved true effect.

fore do not attempt to certify that ARGUS recovers true effects. Instead we construct *local* ground truth by controlled perturbation. Beginning from a credible paper, we inject exactly one known flaw that targets a single identification dimension, drawn from the flaw taxonomy (`flaw_taxonomy.yaml`). Because we control which dimension we damaged, we know the correct audit outcome for that dimension—without ever needing the true effect. The bottom band of Figure 2 shows the loop: the clean and injected versions are each audited, and three metrics are computed.

- **Detection:** does the injected version raise risk on the targeted dimension above a threshold?
- **False alarm:** does the clean version stay quiet on that dimension?
- **Localization:** is the targeted dimension the one the auditor flags most strongly?

4.2 Realistic perturbation, not sentinels

How a flaw is injected is a methodological choice that matters. A naïve injector appends a sentence that *names* the flaw—for instance, “treated and control have non-zero pre-period coefficients with significant leads”. If the detector then searches for exactly those phrases, detection becomes a tautology: the injector plants the very string the detector greps for. In our initial implementation this produced a perfect-looking but meaningless detection rate of 1.000.

Table 1. Synthetic flaw injection (`clean_supported`, `max_steps` = 1, eleven flaws): a keyword baseline, a same-model *single-pass* ablation, and the *two-stage* ARGUS assessor. Keyword and two-stage share the same retrieved evidence; single-pass uses the same model and rubric but removes the architecture. Under matched conditions detection rises from keyword presence, through single-pass adequacy reasoning, to bounded per-dimension auditing. The final row is the leaked, pre-fix circular setup.

Assessor	detection	false alarm	localization
keyword baseline	0.182	0.000	0.182
single-pass LLM (gpt-4o)	0.455	0.000	0.455
two-stage ARGUS (gpt-4o)	1.000	0.000	0.909
<i>(circular, pre-fix)</i>	1.000	0.000	1.000

We replace this with *realistic structural perturbation* of two kinds. *Omission* flaws *remove* the section, figure, or table that supplies the support for a dimension (for example, deleting the placebo-robustness section or the balance table), modelling a paper that simply omits the check; an auditor can flag this non-circularly by noticing the *absence* of evidence. *Commission* flaws *rewrite* a section in flawed-but-natural prose, the way a real flawed paper would describe a mistaken design choice (e.g. pooling all province-years in a two-way fixed-effects regression under staggered timing), *without* containing any of the phrases the detector looks for. We enforce the latter with a regression test that fails if an injected paper contains the target dimension’s negative-signal phrases, so the circularity cannot silently return. Detection is thereby a genuine task rather than a string match, which is what makes the results in §5 meaningful.

5 Preliminary Results

We compare three assessors under the same flaw-injection protocol on the `clean_supported` fixture (one flaw injected per run, all eleven flaws). A deterministic, non-LLM **keyword baseline** scores each dimension from the presence or absence of supporting and risk-indicating vocabulary. The **two-stage ARGUS assessor** judges evidence *adequacy*—whether the reported evidence actually establishes the assumption—over the *same retrieved evidence* as the keyword baseline (extraction is identical), so that pair isolates the judgement policy. A **single-pass** ablation removes the architecture entirely (whole paper and all dimensions in one call; §5.1). Table 1 reports the headline metrics; the per-flaw breakdown is in `experiments/phase1_pilot/`, and Figure 3 is generated from these runs.

The keyword baseline is blind to commission flaws. The baseline never raises a false alarm, and it correctly flags the two omission flaws (`measurement_break` and `spillover_contamination`), whose vocabulary does not appear elsewhere in the paper, detecting them by the *absence* of evidence. But beyond those, detection is only 0.182 (Figure 3, panel B). Two failure modes explain it: *cross-section vocabulary bleed*, where deleting the placebo section does not help because

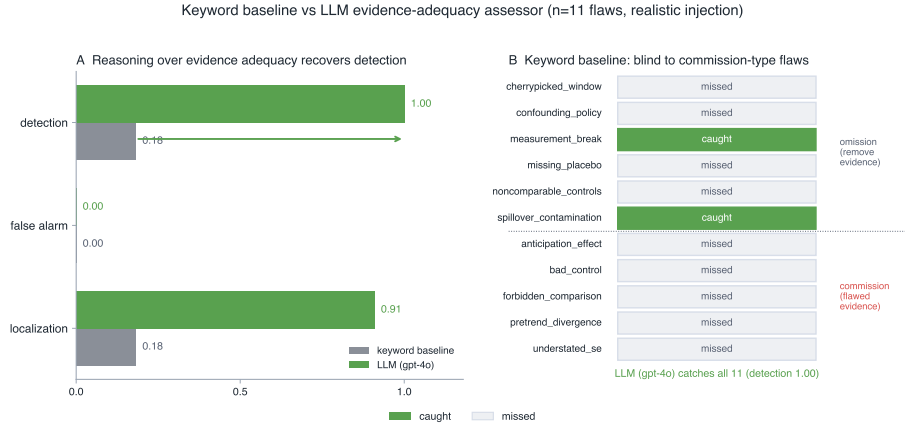


Fig. 3. Keyword baseline vs LLM evidence-adequacy assessor under realistic flaw injection. (A) Swapping only the assessment policy raises detection from 0.182 to 1.000 with false alarm fixed at 0 and localization 0.909. (B) Per-flaw keyword outcomes grouped by injection type: the baseline catches only omission-type threats whose vocabulary is not echoed elsewhere and is blind to every commission-type threat, all of which the LLM catches. Generated by `experiments/phase1_pilot/make_figure2.py` (keyword recomputed live; LLM headline frozen in `llm_summary.json`).

another section also mentions “placebo”; and, more fundamentally, blindness to *every* commission flaw, where flawed-but-plausible evidence is present and uses the expected topic vocabulary. Keyword presence is not evidence adequacy.

Reasoning over adequacy closes the gap. Swapping *only* the assessment policy—extraction is unchanged, so both see the same evidence—to an LLM that judges adequacy raises detection from 0.182 to 1.000 with no false alarms and 0.909 localization. The model reads the flawed-but-natural prose the keyword scorer cannot: for the injected anticipation flaw it returns *high* / *flawed* with the rationale that “outcomes began changing between the policy announcement and rollout,” even though no sentinel keyword is present. This distinction is the empirical core of the project, and it validates the bounded-agentic design of placing agency where reasoning is required.

Scope. These are synthetic fixtures with a small number of dimensions; 1.000 is a proof-of-concept on clearly injected flaws, not a claim of a solved task. The sharper question is whether the result survives on real published papers (§5.2); we first ask what the bounded architecture itself contributes.

5.1 Architecture ablation: single-pass prompting versus bounded auditing

The results above show that an LLM-based assessor can reason about reported evidence adequacy beyond keyword matching, but they do not isolate the contribution of the bounded ARGUS architecture. We therefore add a single-pass ablation: the baseline receives the full paper and the same eleven-dimension rubric in one model call, using the same provider, model (**gpt-4o**), temperature, risk schema, and scoring harness, but removing staged retrieval, relevance gating, bounded tool use, and the explicit evidence-grounding (**unknown**) state. The only difference is architectural.

Synthetic flaws. On the flaw-injection benchmark (Table 1) the single-pass assessor detects only 0.455 of injected flaws (false alarm 0.000, localization 0.455), against 1.000, 0.000, and 0.909 for the bounded two-stage assessor. Judging eleven dimensions from the whole paper at once, the model misses more than half of the planted flaws that per-dimension retrieval and assessment isolate. The architecture therefore contributes to detection itself, not only to how evidence is presented.

Real papers. On the 25 papers common to both runs (275 cells) the contrast is sharper. The single-pass baseline almost never abstains: its **unknown** share is 0.4%, versus 38.9% for the two-stage pipeline. The cells the bounded pipeline marks **unknown**—where the relevance gate could not surface evidence—become confident judgements under single-pass, mostly *low* (1.8% $\rightarrow$ 14.9%) or *medium* (14.2% $\rightarrow$ 43.6%), while the high-risk share is comparable (41.1% vs 45.1%). Without a gate the model commits a substantive risk for nearly every dimension instead of reporting that it could not ground the evidence.

What the architecture adds. The ablation thus attributes a concrete role to the bounded design beyond raw model capability: per-dimension retrieval and assessment recover flaws a single pass misses, and the explicit **unknown** state turns retrieval failure into a visible, auditable signal rather than a confident guess. Separating evidence retrieval from adequacy assessment is what lets ARGUS distinguish substantive identification risk from a grounding failure—the distinction the rest of §5 relies on.

5.2 From synthetic fixtures to real papers: diagnosing the grounding bottleneck

We next audit 27 published DID papers from the CausalVerify corpus [24]—empirical economics broadly, with environmental policy as the motivating case. Here there is no injected ground truth; we report the raw per-dimension risk distribution of each assessor (Figure 4).

A relocated bottleneck on real papers. The keyword baseline is permissive: 91% of its judgements are *low*. On real papers it sees topic vocabulary, declares the dimension supported, and flags almost nothing. The two-stage LLM is differentiated, but its most informative output is that $\sim 39\%$ of judgements are **unknown**: the relevance gate could not surface evidence for that dimension, so the system says so rather than over-flagging. Two regimes appear—dimensions whose discussion is diffuse or lives in figures the text retrieval cannot reach (parallel trends, no anticipation, placebo) are dominated by **unknown**, while dimensions where a section is retrieved (sample period, concurrent policies, SUTVA) are judged inadequate. The finding here is methodological: moving from synthetic fixtures to real papers is bottlenecked not by causal reasoning but by *evidence grounding*. This is why the two-stage design reports retrieval quality and an explicit **unknown** state, distinct from a substantive high-risk judgement (§6).

How the assessor judges evidence. The judgements are grounded in the retrieved text, not pattern-matched. On a trade-and-wages paper ARGUS retrieves the employment-results section and flags *inference* as high because “the paper only mentions heteroskedastic-consistent standard errors and does not provide information on clustering or the number of clusters, which is crucial for the dependence structure in panel data”—a specific econometric observation, not a keyword hit. On another paper it retrieves an “Alternative Market Share Specification” section and flags *robustness/placebo* as high, noting that this “only discusses an alternative specification” and is not a placebo or falsification test. In both cases the model distinguishes a *related but different* reported check from the one the assumption actually requires—now on real papers rather than synthetic fixtures.

Limitations of the real-paper runs. Roughly 45% of the LLM’s judgements on these top-journal papers are *high*; without dimension-level expert labels we cannot say whether this is warranted or over-strict where retrieval succeeds. Calibrating ARGUS against expert annotation on this corpus is the immediate next step (§6); what we claim is the contrast in behaviour and the localization of the grounding bottleneck, not the correctness of any individual flag.

5.3 A calibrated human gold (pilot)

To test whether ARGUS’s flags agree with expert judgement, we built a small expert gold: two annotators independently labelled risk on 5 DID papers $\times$ 11 dimensions (55 cells). Round-1 agreement was low on the raw 3-level scale (Cohen’s $\kappa = 0.17$) but the disagreement is almost entirely ordinal: weighted $\kappa = 0.31$, and 96% of disagreements are a single severity step, so annotators almost never reverse direction. A calibration round then adjudicated the 27 disagreements: 16 resolved to a single label once the low/medium boundary was anchored, and 11 were flagged as intrinsically ambiguous. Those ambiguous cells are not random: they concentrate on *non-canonical DID designs* (continuous-shock, event-study)

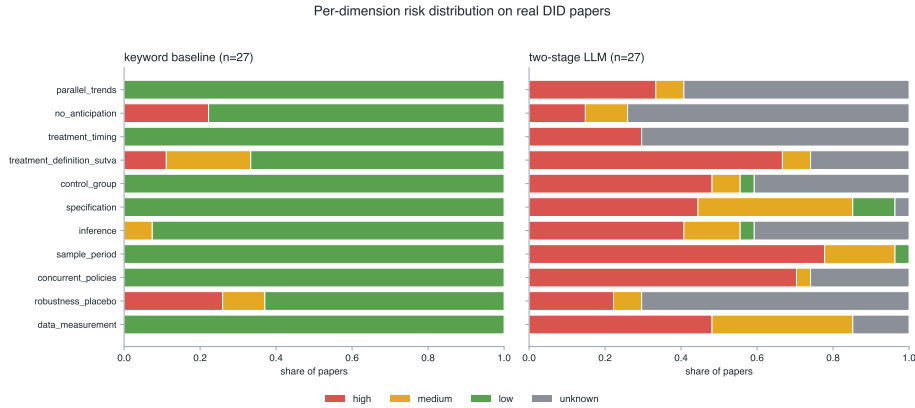


Fig. 4. Per-dimension risk distribution on 27 real DID papers (CausalVerify corpus). The keyword baseline (left) is almost entirely *low*; the two-stage LLM (right) is differentiated, with a large **unknown** share that marks retrieval failure rather than over-flagging. Generated by `experiments/real_papers/make_realcorpus_figure.py` from committed per-paper CSVs.

Table 2. ARGUS-LLM vs adjudicated human gold (55 cells, 5 papers). Two **unknown** policies: coverage (drop abstentions) and strict (**unknown** counts as mismatch). Pilot-scale—only 2 gold cells are *high*, so high precision/recall are noisy; the robust signal is the one-directional over-severity and the abstention rate.

unknown policy	n	scored	exact agr.	Cohen’s κ	weighted κ
exclude (coverage)	33	0.152	0.019	0.044	
mismatch (strict)	55	0.091	0.002	—	

where a dimension like staggered timing is not applicable as conventionally defined. A real share of “expert disagreement” is thus *design applicability*, not severity, which is itself why even experts abstain. The adjudicated gold (55 cells) is low 29, medium 24, high 2. This is a pilot ($n=5$, two annotators, one adjudication round), and we read it as preliminary.

5.4 ARGUS vs the human gold

Comparing ARGUS-LLM’s risk to this gold gives the first calibration signal. Table 2 reports agreement under two **unknown** policies, and Figure 5 the diagnostic breakdown.

Interpretation. The errors are systematic rather than random. ARGUS abstains on 22/55 cells and, when it does answer, its errors are almost entirely one-directional: 28/33 answered cells are more severe than the adjudicated human gold label, while none are less severe. Human gold contains only two high-risk

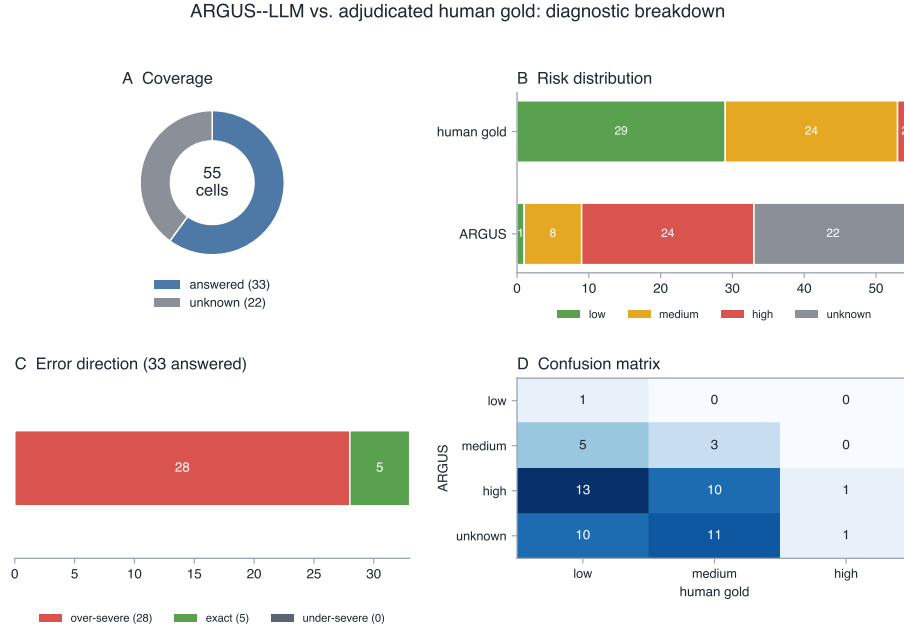


Fig. 5. ARGUS-LLM versus adjudicated human gold on 55 dimension-level audit cells from five papers. Panel A reports coverage, separating answered judgments from abstentions. Panel B compares the severity distributions of human gold and ARGUS outputs. Panel C decomposes answered judgments by error direction among the 33 answered cells. Panel D reports the ARGUS-by-gold confusion matrix. The dominant signal is not random disagreement but one-directional over-severity: ARGUS assigns a higher risk than human gold in 28/33 answered cases and a lower risk in none.

cells, whereas ARGUS assigns high risk to 24 cells. Because the pilot gold set contains very few high labels, high precision and recall should be interpreted cautiously. The more robust conclusion is that the current ARGUS assessor is over-sensitive to missing or partial evidence and is better interpreted as a *high-sensitivity screening tool* for prioritizing human review.

5.5 Calibration: from diagnosis to improvement

The gold comparison also points to a fix. An error analysis of the over-severe cells is sharp: of ARGUS’s over-severe *high* judgements, 23/24 have `evidence_status=missing` and 20/24 have `retrieval_quality=weak`. ARGUS treats “weak retrieval found nothing” as a substantive *high*, conflating a *retrieval failure* with the paper lacking the evidence, which the human, reading the full PDF, resolves to low/medium. This is a fixable calibration target.

We add a deterministic calibration layer over the LLM output, keyed on the rich fields it already produces: the dominant rule converts a weak-retrieval *high*

Table 3. Deterministic calibration vs the human gold (55 cells), two policies for weak-retrieval *high*: demote to *medium* (keep coverage) or abstain to **unknown**. Both cut over-severity and preserve high recall. Pilot scale—only 2 gold cells are *high*, so high precision/recall are noisy.

metric	before	demote→med	abstain→unk
over-severe (of answered)	29	18	6
high precision / recall	0.04 / 0.50	0.50 / 0.50	0.50 / 0.50
exact agreement (answered)	0.12	0.45	0.54
weighted κ (answered)	0.06	0.13	0.25
answered / unknown	33 / 22	33 / 22	13 / 42

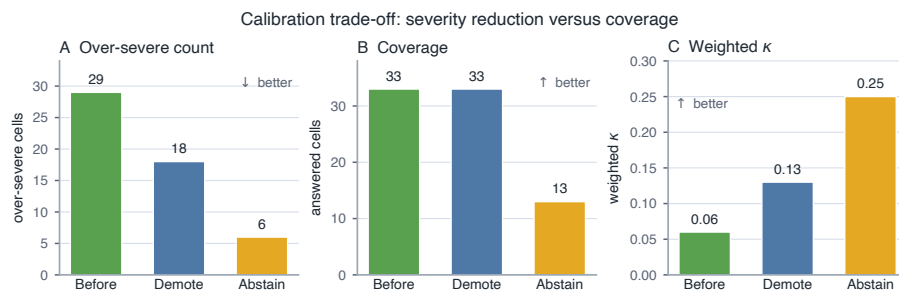


Fig. 6. Calibration trade-off: severity reduction versus coverage against the adjudicated human gold. Demoting weak-retrieval high-risk judgments to *medium* preserves coverage while reducing over-severity. Abstaining instead gives the strongest severity correction and the highest weighted agreement, but sharply reduces coverage. Human-gold calibration can correct severity bias; the remaining bottleneck is evidence grounding rather than severity thresholding alone.

(no contradiction signalled in the rationale) either to *medium* (keep coverage) or to **unknown** (abstain); smaller rules soften a few anachronistic or reporting-gap *high*s. Against the gold (Table 3), both policies cut over-severity and preserve high-risk recall (0.50): severity falls without losing detection. The keep-coverage policy keeps all 33 answers and raises exact agreement from 0.12 to 0.45; the abstain policy reaches the highest weighted κ (0.25) but collapses coverage to 13 answered cells. Neither addresses the root cause, weak retrieval, so the lift that would add *correct* coverage must come from better evidence grounding. The human gold thus does more than evaluate ARGUS; it calibrates it.

Figure 6 visualizes the trade-off behind Table 3. The demotion policy preserves coverage, keeping 33 answered cells, while reducing over-severe judgments from 29 to 18 and improving exact agreement. The abstention policy is more conservative: it reduces over-severe judgments to 6 and improves weighted κ to 0.25, but leaves only 13 answered cells. Thus, calibration can convert erroneous high-risk judgments into either medium-risk judgments or honest abstentions, but it cannot by itself recover the missing evidence needed for correct coverage.

6 Discussion and Research Plan

Threats to validity. These results are preliminary by design. They span synthetic fixtures, 27 real DID papers, and a human gold that is small (5 papers, two annotators, one adjudication round); we read this gold as a preliminary signal, and its two *high* cells make high precision/recall noisy. The keyword baseline is an intentional lower bound. Moreover, flaw injection certifies an auditor against the *injected* threat on a single dimension; it does not certify the absence of other, un-injected threats, and it inherits whatever blind spots the flaw taxonomy has. We therefore treat detection and localization as necessary but not sufficient evidence of an auditor’s quality, and pair them with the false-alarm rate and the human-gold comparison.

Construct validity: what ARGUS measures. A measurement is only as good as the link between the construct and the task that operationalizes it [8,15]. ARGUS audits whether a paper’s identification assumptions are *adequately supported by its reported evidence*—reported-evidence adequacy is an operational proxy for identification credibility, not causal validity itself. The two can diverge: a study may report every diagnostic yet still be invalid, or omit a check yet be sound. ARGUS therefore surfaces risk for an expert to weigh, and a further gap separates the construct from the task: retrieving the relevant evidence. Our real-paper runs show this is the binding constraint. A reasoning assessor handed off-target evidence over-flags every dimension, so ARGUS reports a *retrieval-quality* status and an explicit **unknown** risk (“the system could not find the evidence”) distinct from a substantive high-risk judgement (“the paper lacks it”). The recurring constraint is evidence grounding, not the reasoning itself, which is why ARGUS separates retrieval, relevance filtering, and adequacy assessment.

Research plan. The doctoral programme builds on the result in §5, in four stages. (1) *Harden the reasoning assessor.* The adequacy-reasoning policy already recovers detection on the synthetic fixtures; the next step is to stress it on harder, subtler flaws and to study its calibration, robustness across models, and the traceability of its cited evidence—under the same flaw-injection protocol, so gains stay attributable and reproducible. (2) *Dimension-level expert annotation.* We already audit a real corpus of published DID papers (§5.2); the missing piece is *dimension-level* expert labels of identification risk—the existing corpus is labelled at the method/claim level, not per assumption. Collecting these (with multiple annotators, recording disagreement) would calibrate whether ARGUS’s high/medium/**unknown** judgements match expert opinion and where experts themselves disagree. (3) *Human-expert studies.* Measure whether ARGUS reports actually help domain experts (calibration of trust, time saved, errors avoided), since the system is meant to assist, not replace, expert judgement. (4) *Broader policy domains.* Extend beyond the China emissions-trading case to other environmental and, eventually, general empirical-economics settings, testing how far the rubric-plus-injection methodology generalizes.

7 Conclusion

We have argued that the right target for AI assistance in empirical policy research is not the unobservable causal effect but the *identification credibility* of the study that estimates it. ARGUS operationalizes this stance: a bounded-agentic pipeline decomposes identification into auditable dimensions, gathers and assesses in-paper evidence, and localizes risk in a transparent, human-reviewable report, while confining model agency to two scoped, traceable sub-tasks. We validate such auditors with synthetic flaw injection, which supplies reproducible local ground truth without access to the true effect. Once the leakage that made this evaluation circular is removed, a keyword baseline is blind to flawed-but-plausible evidence (0.182 detection), while a model that judges evidence adequacy recovers it (1.000, no false alarms). The pilot human-gold study further shows that expert labels can do more than evaluate the auditor: they expose a systematic over-severity bias and calibrate the mapping from evidence gaps to audit risk. Scaling this from synthetic fixtures to a real, expert-annotated corpus with humans in the loop is the doctoral programme. Auditing identification credibility is, we argue, both a tractable and a responsible role for AI in empirical environmental-policy research.

References

1. Bertrand, M., Duflo, E., Mullainathan, S.: How much should we trust differences-in-differences estimates? *Quarterly Journal of Economics* **119**(1), 249–275 (2004)
2. Botvinik-Nezer, R., Holzmeister, F., Camerer, C.F., et al.: Variability in the analysis of a single neuroimaging dataset by many teams. *Nature* **582**(7810), 84–88 (2020)
3. Brodeur, A., Cook, N., Heyes, A.: Methods matter: P-hacking and publication bias in causal analysis in economics. *American Economic Review* **110**(11), 3634–3660 (2020)
4. Callaway, B., Sant’Anna, P.H.C.: Difference-in-differences with multiple time periods. *Journal of Econometrics* **225**(2), 200–230 (2021)
5. de Chaisemartin, C., D’Haultfœuille, X.: Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review* **110**(9), 2964–2996 (2020)
6. Geburu, T., Morgenstern, J., Vecchione, B., Vaughan, J.W., Wallach, H., Daumé III, H., Crawford, K.: Datasheets for datasets. *Communications of the ACM* **64**(12), 86–92 (2021)
7. Goodman-Bacon, A.: Difference-in-differences with variation in treatment timing. *Journal of Econometrics* **225**(2), 254–277 (2021)
8. Jacobs, A.Z., Wallach, H.: Measurement and fairness. In: *ACM Conference on Fairness, Accountability, and Transparency (FAccT)* (2021)
9. Jin, Z., Liu, J., Lyu, Z., Poff, S., Sachan, M., Mihalcea, R., Diab, M., Schölkopf, B.: Can large language models infer causation from correlation? In: *International Conference on Learning Representations (ICLR)* (2024)
10. Jin, Z., et al.: CLadder: Assessing causal reasoning in language models. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2023)

11. Kiciman, E., Ness, R., Sharma, A., Tan, C.: Causal reasoning and large language models: Opening a new frontier for causality. *Transactions on Machine Learning Research (TMLR)* (2024)
12. Kiela, D., Bartolo, M., Nie, Y., et al.: Dynabench: Rethinking benchmarking in NLP. In: *NAACL-HLT* (2021)
13. Lee, D., Yun, H., Cha, M., Park, S., Park, S., Kim, J.: EconCausal: A context-aware causal reasoning benchmark for large language models in social science. Preprint, arXiv:2510.07231 (2025)
14. Liang, P., Bommasani, R., Lee, T., et al.: Holistic evaluation of language models. *Transactions on Machine Learning Research (TMLR)* (2023)
15. Raji, I.D., Bender, E.M., Paullada, A., Denton, E., Hanna, A.: AI and the everything in the whole wide world benchmark. In: *NeurIPS Datasets and Benchmarks Track* (2021)
16. Roth, J., Sant’Anna, P.H.C., Bilinski, A., Poe, J.: What’s trending in difference-in-differences? a synthesis of the recent econometrics literature. *Journal of Econometrics* **235**(2), 2218–2244 (2023)
17. Sawarni, A., Tan, J., Syrgkanis, V.: CausalReasoningBenchmark: A real-world benchmark for disentangled evaluation of causal identification and estimation. Preprint, arXiv:2602.20571 (2026)
18. Silberzahn, R., Uhlmann, E.L., Martin, D.P., et al.: Many analysts, one data set: Making transparent how variations in analytic choices affect results. *Advances in Methods and Practices in Psychological Science* **1**(3), 337–356 (2018)
19. Sun, L., Abraham, S.: Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics* **225**(2), 175–199 (2021)
20. Yang, L., Shirvaikar, V., Clivio, O., Falck, F.: A critical review of causal reasoning benchmarks for large language models. Preprint, arXiv:2407.08029 (2024)
21. Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., Cao, Y.: ReAct: Synergizing reasoning and acting in language models. In: *International Conference on Learning Representations (ICLR)* (2023)
22. Zečević, M., Willig, M., Dhami, D.S., Kersting, K.: Causal parrots: Large language models may talk causality but are not causal. *Transactions on Machine Learning Research (TMLR)* (2023)
23. Zhang, Y., et al.: CausalVerify: An execution-grounded benchmark for LLM causal-inference workflows (2026), manuscript under review
24. Zhang, Y., et al.: CausalVerify: corpus of published economics papers and synthetic dgps. Hugging Face dataset, <https://huggingface.co/datasets/causalverify/causalverify-neurips2026> (2026), tag `neurips2026-submission`
25. Zhou, Y., Wu, X., Huang, B., Wu, J., Feng, L., Tan, K.C.: CausalBench: A comprehensive benchmark for causal learning capability of large language models. Preprint, arXiv:2404.06349 (2024)